

Day 07

1/12/2025

→ While loop.

It is used when you do not know in advance how many times the loop needs to run.

* It continues executing until the condition becomes false.

ex:- $i \geq 1$

```
while  $i \leq 5$ :
```

```
    print(i)
```

```
    i += 1
```

1) What are the loop statements?

⇒ Loop statements are "control structures" that allow a set of instructions to be executed repeatedly until a certain condition is met.

Types of loop statement.

1. Break.

2. Continue

3. Pass

2. Break:- used to exit / terminate the loop immediately

ex:- for i in range(1, 11, 1):

```
    if  $(i == 5)$ :
```

```
        break
```

```
    else:
```

```
        print(i)
```

output: 1

2

3

4

2. Continue Statements :-

It is used to skip the current iteration of the loop

ex:- for i in range(1, 11, 1):

if (i == 5):

continue

else:

print(i).

3. Pass Statement :-

It purely does nothing.

ex:- for i in range(1, 11, 1):

if (i == 5):

pass

else:

print(i)

Example problem.

Write a program to check the given number is prime number or not.

```
=> num = int(input("enter a number"))
```

```
for i in range(2, num, 1):
```

```
if num % i == 0:
```

```
    print(num, "it's not a prime no")
```

```
else:
```

```
    print(num, "it's a prime no")
```

Q27 Write a program to count no of natural no,
even no, odd no from the given range.

```
num = int(input("enter a number"))  
num2 = int(input("enter a number"))
```

```
natural = 0
```

```
even = 0
```

```
odd = 0
```

```
for i in range(num, num2+1, 1):  
    natural = natural + 1
```

```
    if i % 2 == 0:  
        even = even + 1
```

```
    else:
```

```
        odd = odd + 1
```

```
print("natural no =", natural)
```

```
print("even number =", even)
```

```
print("odd number =", odd)
```

Python Condition (Basic)

1) even @ odd = if num % 2 == 0

2) + @ - = if num > 0
elif num < 0
else.

3) longest of 2 no = if a > b.

4) Print natural no from 1 to n = for i in range(1, n+1)

5) Sum of first n natural no = sum = sum + i

6) Factorial no = fact = fact * i

7) print multiplication table = for i in range(1, 11)

8) prime no = if n % i != 0

9) Reverse a no = rev = rev * 10 + rem
(palindrome) rem = n % 10

10) Sum of digits = rem = n % 10
n = n // 10

11) Count digit = while n > 0:
count += 1

12) Check no is Armstrong = sum + = digit ** no-of-digit

13) generate fibonacci series = next = a+b

14) check palindrome = if num == rev.
(Reverse)

15) find HCF of 2 no = while b != 0:
a, b = b, a % b

16) find LCM of 2 no = Lcm = (a*b) // gcd

17) print all prime no = for i in range(2, n):

18) Check Armstrong no = sum + = factorial(digit)

19) perfect no check = if sum == n:

20) print pattern programs = ex: triangle pattern,
pyramid

= for i in range(1):

= for j in range(1):

21) Harshad no = num % (sum of digit) == 0

22) second largest in list = sorted(list)

23) Remove duplicate = set(list)

24) Check string is anagram = if sorted(s1)
== sorted(s2)